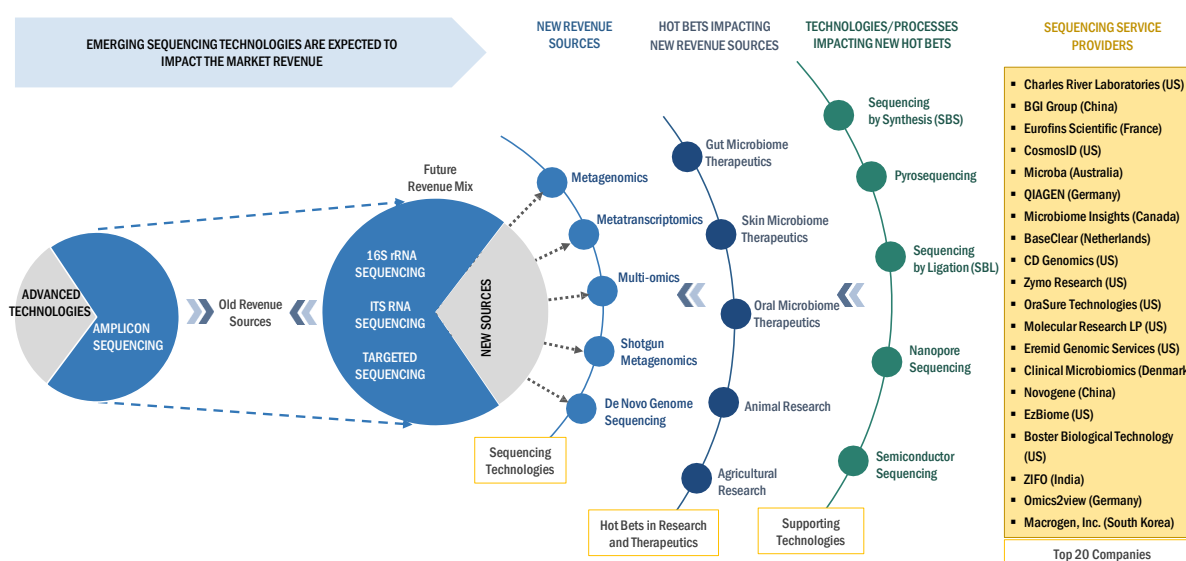


5.4 TRENDS AND DISRUPTIONS IMPACTING CUSTOMERS' BUSINESSES

Robust research activities to explore the potential of the human microbiome present great potential in a number of treatments against obesity, inflammatory bowel disease, and diabetes, among other conditions. Information on microbiomes helps in the identification of the causal links between the microbiome and disease, further facilitating the development of efficient and targeted treatment strategies. Additionally, progress in state-of-the-art sequencing approaches, computational analyses, and experimental assays is supporting the growing demand for microbiome sequencing.

FIGURE 20 REVENUE SHIFT AND NEW REVENUE POCKETS FOR MICROBIOME SEQUENCING SERVICE PROVIDERS



Source: MarketsandMarkets analysis

The most commonly used sequencing technologies include 16S rRNA sequencing, ITS RNA sequencing, and whole-genome sequencing. Sequencing and experimental analyses of the microbiome continue to advance substantially. By coupling multi-omics technologies, statistical and computational analyses, and more advanced disease models, these approaches promisingly provide the opportunity to establish disease causality and subsequently inform therapeutic development.

5.5 VALUE CHAIN ANALYSIS

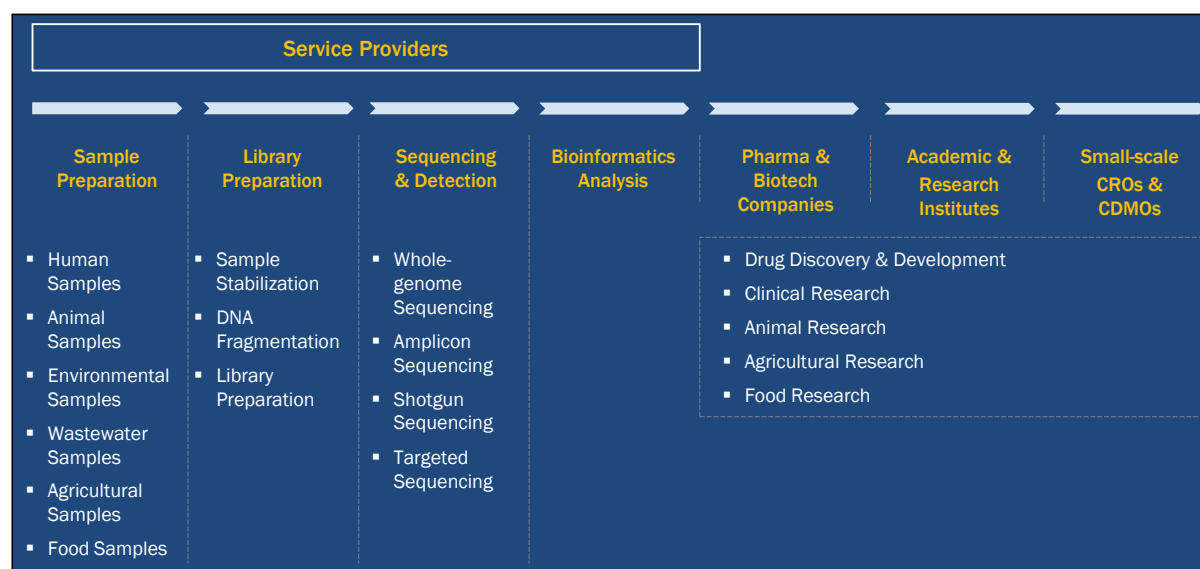
SAMPLE PREPARATION AND STUDY DESIGN SERVICES

Microbiome sequencing services start with outsourcing of study design, wherein the service provider helps in the implementation and validation of the hypothesis and the variables of interest in a sequencing project. Different variables in the aspect of study design include the number of samples required, apt sequencing technology to be used, and the requirement of follow-up studies. Service providers have expertise with respect to the collection, storage, and transport options optimized for this workflow.

SEQUENCING, LIBRARY PREPARATION, AND DATA ANALYSIS SERVICES

NGS instruments are predominantly used for the sequencing of DNA and RNA samples. Shotgun metagenomics sequencing is offered by a majority of sequencing service providers, even for deep-sequenced samples. DNA library preparation is verified for quality and yield. Genomic DNA is fragmented at specific sizes suitable for the purification/separation of desired proteins and nucleic acids. The selected fragments are then polished, A-tailed, and ligated with the full-length adapter. NGS technology is employed to sequence the sample, and the final stage involves bioinformatics analysis.

FIGURE 21 VALUE CHAIN ANALYSIS OF MICROBIOME SEQUENCING SERVICES MARKET



Source: MarketsandMarkets Analysis